

Part 1: Bayesian Reasoning

1. Prior probability is 0.003

2. Likelihood is $P(\text{cancer} \wedge \text{+ve test result}) + P(\text{no cancer} \wedge \text{+ve test result})$ as that's how $P(\text{+ve test result})$ is calculated; so likelihood is $P(\text{cancer} \wedge \text{+ve test result}) + P(\text{no cancer} \wedge \text{+ve test result}) = P(\text{cancer})P(\text{+ve test result}|\text{cancer}) + P(\text{no cancer})P(\text{+ve test result}|\text{no cancer})$
so likelihood is $P(\text{cancer})P(\text{+ve test result}|\text{cancer}) + P(\text{no cancer})P(\text{+ve test result}|\text{no cancer}) = (0.3\%)(50\%) + [1 - P(\text{cancer})](3\%) = (0.3\%)(50\%) + [1 - (0.3\%)](3\%) = 0.3\%(50\%) + 0.997(3\%)$
so likelihood is $(0.3\%)(50\%) + [1 - (0.3\%)](3\%) = 3.141\%$

3. The observation given was a positive test result

4. $P(\text{cancer}|\text{+ve test result}) = \frac{P(\text{+ve test result}|\text{cancer})P(\text{cancer})}{P(\text{+ve test result})} = \frac{(50\%)(0.003)}{(3.141\%)} = \frac{50}{1047}$, so this is the posterior

5. $P(\text{no cancer}|\text{+ve test result}) = 1 - P(\text{cancer}|\text{+ve test result}) = 1 - \frac{50}{1047} = \frac{997}{1047}$

Part 2: Frequencies

1. Out of people with a positive test result, 15 have cancer out of 314 people.
2. The part 2 solution is easier. Probability isn't intuitive to me, I don't understand it well. I have trouble with it. But frequencies of people having things are easier for me to get.
3. I'd try to tell them that out of 314 people with positive results, only 15 people have cancer; that means 299 of them don't. This person shouldn't be too scared since most people getting positive tests don't have cancer.